

MUNICTECH ENTERPRISE DEPLOYMENT & EUROPE BRIEF — FINAL

AI Action Path Assurance by HAIEC

WHY THIS MATTERS TO ENTERPRISES

AI agents are becoming operators.

They can search internal systems, update records, send messages, change configurations, invoke cloud services, interact with APIs, create persistent state, and coordinate with other agents.

Enterprises already have tools for code scanning, IAM, observability, governance, and runtime protection.

The missing question is often:

What source-backed path connects this AI-facing capability to a real business or system consequence, and does that path align with the authority the organization intended?

AI Action Path Assurance is designed to answer that question.

WHERE HAIEC SITS

Repository / AI system

- HAIEC source and capability intelligence
- AI Action Path Trail
- authority and control reconciliation
- assurance decision
- release evidence
- reports / System Consequence Map / machine output / Passport / Decision Receipt.

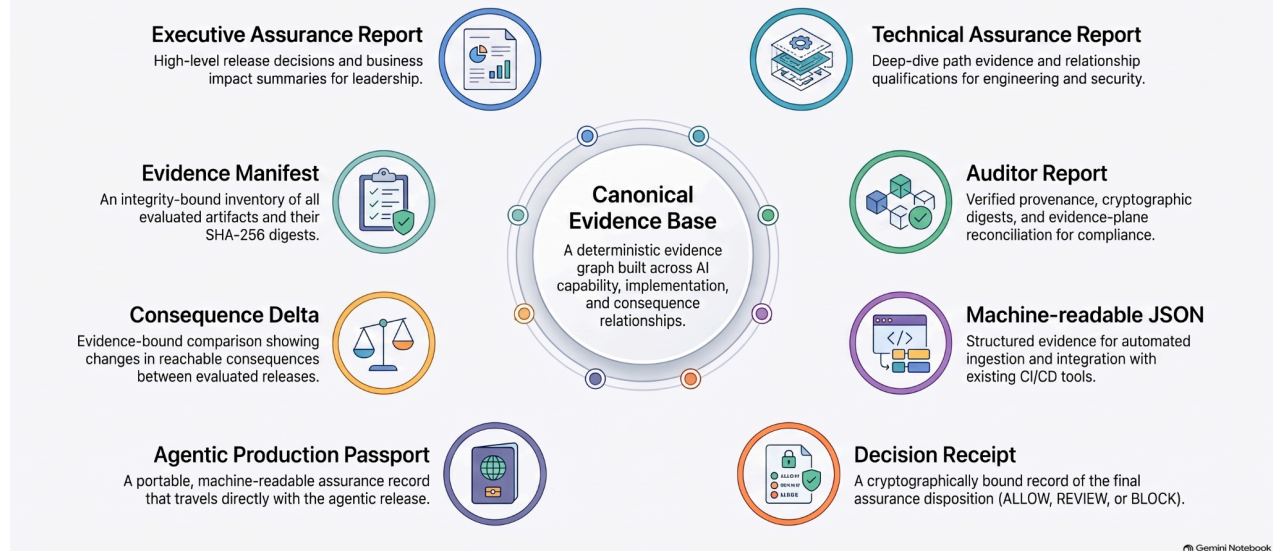
HAIEC is an assurance layer. It complements AppSec, IAM, runtime monitoring, policy systems, and governance platforms rather than replacing them.

PORTABLE EVIDENCE FOR DIFFERENT AUDIENCES

HAIEC composes one canonical evidence base and projects it into audience-specific outputs. The underlying assurance result is not reinterpreted for each audience.

Different audience. Same evidence.

Portable evidence that can travel with the release.



One evidence base can support executive, technical, auditor, machine-readable, receipt, passport, delta, and manifest views while preserving the same evaluated truth.

ENTERPRISE ADOPTION PATH

1. START WITH THE STANDARD MODEL

Connect or evaluate the target source snapshot.

HAIEC identifies supported AI-facing capabilities, traces qualified paths through implementation code, and maps consequential operations using its standard capability model.

The enterprise does not have to define every action before the first evaluation.

2. ADD ENTERPRISE-SPECIFIC CAPABILITIES ONLY WHERE NEEDED

Some organizations expose proprietary operations that are not part of a generic public API catalog.

Examples:

```
Telecom
restartCell()
changeAntennaTilt()
publishRAppPolicy()
reconfigureCarrier()
```

Banking

releaseWireTransfer()

freezeAccount()

adjustCreditLimit()

Manufacturing

restartProductionCell()

changeRobotSpeed()

publishPLCConfiguration()

These actions can be represented through qualified enterprise capability packs so HAIEC understands the technical action and its business consequence.

3. DEFINE THE APPROVED OPERATING ENVELOPE

The enterprise can specify the boundary inside which the AI-enabled system is intended to operate.

Examples of envelope dimensions:

allowed operations,

resource scopes,

data classes,

destinations,

approved models,

approved tools,

allowed environments,

target-count limits,

change-magnitude limits,

approval requirements.

This converts business intent into an explicit technical comparison boundary.

4. RECONCILE CAPABILITY WITH AUTHORITY

HAIEC keeps five evidence questions independent:

Requested

What was the system intended or requested to do?

Policy Authorized

What did policy approve?

Effectively Granted

What authority was actually established through credentials, configuration, or platform state?

Code Capable

What consequential operation can the evaluated source reach?

Observed

What qualifying runtime evidence shows actually happened?

The value is the comparison.

A system may be Code Capable but not yet have evidence of approved authority.

A policy may authorize an action that the AI-facing code cannot reach.

A runtime event may exist without the source or policy evidence needed for a broader conclusion.

5. RELEASE WITH PORTABLE EVIDENCE

The evaluated result can be projected into:

executive report,

technical report,

auditor / evidence report,

System Consequence Map,

machine-readable evidence,

Agentic Production Passport,

Decision Receipt.

The evidence remains bound to the evaluated release context.

6. COMPARE THE NEXT RELEASE

Consequence Delta shows how the consequence and control surface changes between qualified evaluations. In the MunichTech A/B-8 comparison, HAIEC established 12 CONTROL_CHANGED items while leaving 99 positional facts UNRESOLVED rather than manufacturing addition/removal claims where absence was not proven.

This turns assurance into a release workflow rather than a one-time assessment.

BUYERS AND USERS

AI Platform Engineering

Needs to understand what agentic systems can actually reach before release.

Application Security

Needs code-grounded action evidence beyond generic findings.

Security Architecture / Cloud Security

Needs to understand service, resource, credential, egress, and authority relationships.

AI Governance / Assurance

Needs technical evidence that can support internal policy and governance decisions.

Release Management

Needs a clear decision record and visibility into consequence-changing releases.

Internal Audit / Independent Assurance

Needs inspectable evidence, provenance, and repeatable outputs.

WHY EUROPE

MunichTech asks teams to build usable, deployable technology with relevance to Europe's competitiveness and strategic technology needs.

Europe's AI challenge is not only model development.

It is also deploying autonomous and semi-autonomous systems into telecom, manufacturing, finance, infrastructure, public services, healthcare operations, logistics, and enterprise software while preserving accountability.

AI Action Path Assurance supports that need through:

PROVIDER-INDEPENDENT ASSURANCE

The assurance question is about capability, authority, consequence, and evidence, not loyalty to one AI model or cloud provider.

INSPECTABLE TECHNICAL ACCOUNTABILITY

Evidence is tied to source-backed relationships and bounded claims rather than another opaque AI judgment.

ENTERPRISE CONTROL

Organizations can define their own approved operating boundaries and proprietary actions.

RELEASE-LEVEL GOVERNANCE

Evidence can travel with a specific evaluated release and be compared over time.

DIGITAL SOVEREIGNTY SUPPORT

Enterprises can retain direct visibility into how their own AI-enabled software reaches internal systems and consequential actions instead of relying solely on external black-box assessments.

INDUSTRY 4.0 RELEVANCE

Manufacturing and industrial automation increasingly combine AI reasoning with systems that can change real operational state.

TELECOM EXAMPLE

Imagine an AI-assisted network automation application.

Standard discovery may identify:

agent,
tool exposure,
service calls,
configuration writes,
authentication context,
approval controls,
resource scope.

The telecom operator may then add proprietary operations such as:

restartCell()
changeAntennaTilt()
publishRAppPolicy()

The approved Operating Envelope might say:

- only selected regions;
- maximum five cells per action;
- production changes require human approval;
- only approved rApps and tools;
- specific change-magnitude limits.

HAIEC can then compare what the source makes reachable against that enterprise-defined boundary.

This is more useful than only asking:

“Does the service account have permission?”

BANKING EXAMPLE

An AI workflow may have access to account and payment APIs.

Standard analysis can identify API exposure and consequential operations.

A bank can add proprietary business actions such as:

releaseWireTransfer()
freezeAccount()

adjustCreditLimit()

The relevant assurance question becomes:

Can the AI reach this operation?

What account or resource scope is bound?

What approval stage is on the path?

What did policy authorize?

What authority is actually granted?

What was observed?

MANUFACTURING EXAMPLE

An AI agent may interact with production scheduling, robotics, or control-plane services.

A manufacturer can extend the capability model with:

restartProductionCell()

changeRobotSpeed()

publishPLCConfiguration()

The Operating Envelope can then encode plant, line, environment, target-count, and approval boundaries.

DEPLOYMENT MODEL

Current:

HAIEC operates as a SaaS with organization-scoped tenancy and evaluates frozen source snapshots to produce evidence-bound assurance artifacts.

Near-term enterprise deployment paths:

- design-partner evaluations;
- CI / release-gate integration;
- proprietary capability packs;
- runtime evidence connectors where Observed proof is required;
- baseline / candidate Consequence Delta for release comparison;
- integration into security, governance, and audit workflows.

WHAT HAIEC DOES NOT REQUIRE

HAIEC does not require the enterprise to:

- replace IAM;
- replace runtime protection;
- manually define every common capability;
- trust an AI model's unsupported opinion;
- treat every action path as a vulnerability.

WHAT HAIEC ADDS

A consequence-centered assurance layer that answers:

What can the AI-enabled code reach?
What real action can that path cause?
What authority and controls apply?
What has actually been observed?
What changed in the next release?
What evidence travels with the decision?

That is the enterprise value of AI Action Path Assurance

SOURCE-CODE TRUST MODEL

The enterprise trust answer should be accurate and strong:
Your source is processed, not retained.

Normal SaaS workflow:

1. Customer connects GitHub account through OAuth.
2. Customer authorizes repository access through the GitHub App installation model.
3. HAIEC evaluates only the authorized repository scope.
4. Source is processed transiently in an ephemeral analysis environment.
5. The analysis environment is discarded after processing.
6. HAIEC retains structured Action Path evidence, findings, hashes, metadata, assurance decisions and release artifacts.
7. HAIEC does not retain a persisted repository copy as the analysis dataset.

This is stronger than saying “we never see code.” HAIEC must temporarily access source to analyze it. The security point is controlled access, transient processing and evidence-only retention.

ENTERPRISE DEPLOYMENT PATTERNS

Current pattern: HAIEC Cloud / SaaS

Best for teams that can authorize repository analysis through GitHub and want fast release-review evidence.

Enterprise PoC pattern: customer-controlled analyzer

For organizations with stricter source-handling requirements, HAIEC can scope a private PoC where analysis runs in a customer-controlled or isolated environment and only structured evidence is exported to the assurance workflow.

Future sovereign pattern: self-hosted / VPC deployment

For telecom, banking, public sector, critical infrastructure or highly regulated European deployments, a hardened self-hosted or VPC architecture is the natural productization path. Do not claim this is generally available until it is built and validated.

MONDAY-MORNING ENTERPRISE FLOW

1. Connect repository or provide an approved source snapshot.
2. Run standard capability recognition.
3. Inspect the System Consequence Map.
4. Select high-impact Action Path Trails.
5. Add proprietary capability definitions only where standard recognition needs domain context.
6. Define or import the approved Operating Envelope.
7. Resolve release review flags by connecting intent, policy, effective authority, control and runtime evidence.
8. Re-evaluate to move from REVIEW toward ALLOW or BLOCK.
9. Carry the release evidence forward through report, Passport, Decision Receipt and machine-readable output.
10. Compare the next release with Consequence Delta.

WHAT BUYERS GET

Faster release review: less manual reconstruction of agent → tool → code → consequence paths.

Better change visibility: clear view of whether a release expanded, narrowed or changed consequential AI reach.

Reusable assurance evidence: one evaluated evidence base can serve engineering, security, governance, release management and audit.

Enterprise extensibility: standard model first, proprietary capability packs only where domain-specific operations require them.

MEASUREMENT DISCIPLINE

Do not publish unsupported claims about scan time, maximum repository size, precision, recall, language coverage, certification status or full on-prem availability. Add those only when supported by measured benchmark or validated product evidence.

.